

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – 7

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Class: TYIT	Batch: 1
Date of Assignment: 4/08/2026	Date/Time of Submission: 4/08/2026

A) Develop a PWA that works in offline mode using Service Worker.

Code:

Offline.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Offline</title>
</head>
<body>
<header>
  <h2>My College Website</h2>
</header>
<main>
  <h2> You are Offline</h2>
  <p>Please check your internet connection.</p>
  <button onclick="location.reload()">Try
Again</button>
</main>
<footer>
  <p>© 2026 My College Website</p>
</footer>
</body>
</html>
```

Service-worker.json:

```
const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/offline.html"
      ]);
    }) );
  console.log("Service Worker Installed");
});
```

```
});
self.addEventListener("fetch", e => {
  e.respondWith(
    fetch(e.request).catch(() =>
      caches.match("/offline.html"))
  );
});
```

Add to _Layout.cshtml

Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

Before the closing </body> tag:

```
<script>
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/service-worker.js');
}
</script>
```

Output:



B) Create a Student Record Management PWA.**Code:****service-worker.js:-**

```
const CACHE = "v1";
self.addEventListener("install", e => {
  e.waitUntil(
    caches.open(CACHE).then(cache => {
      return cache.addAll([
        "/",
        "/Home/Index",
        "/css/site.css",
        "/js/site.js"
      ]);
    })
  );
});
self.addEventListener("fetch", e => {
  e.respondWith(
    caches.match(e.request).then(response => {
      return response || fetch(e.request);
    })
  );
});
```

manifest.json:-

```
{
  "name": "college portal",
  "short_name": "cp",
  "start_url": "/",
  "display": "standalone",
  "icons": [
    {
      "src": "icons/icon1.png",
      "sizes": "148x148",
      "type": "image/png",
      "purpose": "any"
    },
    {
      "src": "icons/icon2.png",
      "sizes": "512x512",
```

```
"type": "image/png",
"purpose": "any"
} ] }
```

Homecontroller.cs:-

```
public IActionResult Index()
{
  student s = new student();
  s.RollNo = 17;
  s.Name = "Radhika";
  return View(s);
}
```

Index.cshtml:-

```
<h1>Offline Page</h1>
<h2>Student Record</h2>
<p>Roll No : @Model.RollNo</p>
<p>Name : @Model.Name</p>
```

Student.cs:-

```
namespace WebApplication12.Models
{
  public class student
  {
    public int RollNo { get; set; }
    public string Name { get; set; } } }
```

Add to _Layout.cshtml

Inside the <head> section:

```
<link rel="manifest" href="/manifest.json">
```

Before the closing </body> tag:

```
<script>
if ('serviceWorker' in navigator) {
  navigator.serviceWorker.register('/service-worker.js');
}
</script>
```

Output: